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Practice Test

Advance Programming

1/5

1. What information will the following code output?

```
import pandas as pd

s = pd.Series([100, 200, 300, 400], index=['Q1', 'Q2', 'Q3', 'Q4'])

print(s.iloc[1:3])
```

☒ Q2 200, Q3 300

iloc[] uses integer position-based slicing, which is exclusive of the end index. So positions 1 and 2 correspond to Q2 and Q3

☐ Q1 100, Q2 200, Q3 300

Know Reason ▼

☐ Q1 100, Q2 200

Know Reason ▼

2. Study the quarterly sales data as follows

```
import pandas as pd

df = pd.DataFrame({
    'Region': ['East', 'West', 'East', 'North', 'West'],
    'Sales': [1200, None, 1700, 1300, 1500]
})

print(df.groupby('Region')['Sales'].sum())
What does this return?
```

☒ Ignores NaN and sums remaining values

sum() in Pandas skips NaN by default. West will have 1500 (ignoring None).

3. Consider employee data as follows

```
import pandas as pd

df = pd.DataFrame({
    'TDS': [15, 25, 20],
    'Salary': [45000, 60000, 52000]
})

print(df.apply(lambda x: x['TDS'] * x['Salary'] / 100, axis = 0))
```

What will the first row of output be?



No output because of key error



When axis=0 is specified, Pandas performs operations across rows. Since 'TDS' and 'Salary' are not row indices, this will result in a KeyError and no output will be produced.

4. Which statements are correct? [More than one option may be correct]



groupby() can group by multiple columns



groupby() can group by multiple columns and does not modify in place. agg() can apply multiple functions to multiple columns.



.agg() can apply multiple functions to multiple columns



groupby() can group by multiple columns and does not modify in place. agg() can apply multiple functions to multiple columns.

5. Which statements are correct? [More than one option may be correct]



A table can have multiple UNIQUE constraints but only one PRIMARY KEY



FOREIGN KEY must reference a UNIQUE or PRIMARY KEY column. PRIMARY KEY cannot have NULLs. CHECK can involve multiple columns (e.g., salary > bonus).



FOREIGN KEY can reference a column that is not UNIQUE

Know Reason ▼



CHECK can validate conditions across multiple columns



FOREIGN KEY must reference a UNIQUE or PRIMARY KEY column. PRIMARY KEY cannot have NULLs. CHECK can involve multiple columns (e.g., salary > bonus).

EDA

1/5

1. Which of the following is true for a perfectly normal distribution?

☐ Mean, median, and mode are equal

Know Reason ▼

☐ Skewness = 0 and kurtosis = 3

Know Reason ▼

☐ 95% of data lies within ± 2 standard deviations of the mean

Know Reason ▼

☒ All of the above

^

A normal distribution is symmetric with mean = median = mode, skewness = 0, and kurtosis = 3. Also, according to the empirical rule, around 95% of the data falls within ± 2 standard deviations of the mean.

2. Which formula correctly represents the Z-statistic for a one-sample Z-test?

☒ $((\bar{x} - \mu)/\sigma)$

Know Reason ▼

☒ $((\bar{x} - \mu)/(\sigma/\sqrt{n}))$

^

$$Z = \frac{\text{sample mean} - \text{population mean}}{\text{standard error}}$$
$$\text{Standard error} = \frac{\sigma}{\sqrt{n}}$$

Assumes population σ is known and the sample size is large

☐ $((\mu - \bar{x})/n)$

Know Reason ▼

3. Which statement is correct?

☐ Type I error occurs when a true null hypothesis is rejected

Know Reason ▼

☐ Type II error occurs when a false null hypothesis is accepted

Know Reason ▼

☐ Increasing sample size reduces Type II error

Know Reason ▼

☒ All of the above

^

Type I = false positive, Type II = false negative. Larger sample size reduces β (Type II error) because test power increases.

4. Which statements are true? [More than one option may be correct]

☒ Histogram shows frequency distribution of numeric data

^

All statements are correct: histogram for numeric frequency, boxplot for spread outliers, scatter plot for numeric correlation, bar chart for categorical comparison.

☒ Boxplot displays median, quartiles, and potential outliers

^

All statements are correct: histogram for numeric frequency, boxplot for spread outliers, scatter plot for numeric correlation, bar chart for categorical comparison.

☒ Scatter plot is ideal for visualizing correlation between two numeric variables

^

All statements are correct: histogram for numeric frequency, boxplot for spread outliers, scatter plot for numeric correlation, bar chart for categorical comparison.

5. Which statements about Z-test are correct?

5/5

☒ Used when population variance is known



The Z-test assumes a known population variance (σ) and generally requires a large sample size (n). For small samples, population normality is important, while for larger samples, the Central Limit Theorem allows normal approximation.

☒ Requires large sample size ($n \geq 30$)



The Z-test assumes a known population variance (σ) and generally requires a large sample size (n). For small samples, population normality is important, while for larger samples, the Central Limit Theorem allows normal approximation.

☐ Assumes population is approximately normal for small samples



The Z-test assumes a known population variance (σ) and generally requires a large sample size (n). For small samples, population normality is important, while for larger samples, the Central Limit Theorem allows normal approximation.

☐ Can be applied when population is not normal if n is large (CLT)



The Z-test assumes a known population variance (σ) and generally requires a large sample size (n). For small samples, population normality is important, while for larger samples, the Central Limit Theorem allows normal approximation.

Cloud Computing and BigData

1. Which PySpark function is commonly used to read a CSV file into a DataFrame?

☒ `spark.read.csv()`



PySpark uses `spark.read.csv()` to load CSV files into DataFrames

2. Given a valid data frame and PySpark environment, what does the following PySpark code return?
`df.select("age").distinct().count()`

☐ Total number of rows in DataFrame

Know Reason ▼

☒ Number of unique values in column age

`distinct()` removes duplicates, and `count()` returns the number of unique entries in age

3. Which command registers a PySpark DataFrame as a temporary SQL table?

☒ `df.createOrReplaceTempView("table_name")`

`df.createOrReplaceTempView("table_name")` command registers a DataFrame as a temporary SQL view so that SQL queries can be executed directly using Spark SQL.

4. Which command builds a Docker image from a Dockerfile?

☒ `docker build -t image_name .`

The `docker build` command creates an image from a Dockerfile, where `-t` assigns a name to the image, and the `.` specifies the build context.

5. Which PySpark function computes summary statistics like mean and standard deviation for numeric columns?

☒ `df.describe()`

The `describe()` function in PySpark generates summary statistics such as count, mean, standard deviation, min, and max for numeric columns.

6. In FastAPI, which decorator is used to define a GET endpoint?

☒ `@app.get()`

FastAPI uses decorators like `@app.get()` to define GET endpoints, allowing API clients to retrieve data from the server, etc

7. Which commands are used to stage and commit changes in Git? [More than one option may be correct]

☒ git add <file>



git add stages changes, and git commit saves them locally. git push uploads to remote, git init initializes a repo.

☒ git commit -m "message"



git add stages changes, and git commit saves them locally. git push uploads to remote, git init initializes a repo.

8. Which statements about cloud computing are correct? [More than one option may be correct]

☒ It provides on-demand resources



Cloud computing provides on-demand resource availability, reduces reliance on physical servers, and enables scalability, though zero downtime is not guaranteed.

☐ It eliminates the need for physical servers



Cloud computing provides on-demand resource availability, reduces reliance on physical servers, and enables scalability, though zero downtime is not guaranteed.

☐ It guarantees zero downtime

Know Reason

☒ It supports scalability



Cloud computing provides on-demand resource availability, reduces reliance on physical servers, and enables scalability, though zero downtime is not guaranteed.

9. Which features are provided by Pydantic models? [More than one option may be correct]

☒ Data validation



Pydantic validates data, converts types, and supports JSON serialisation. It does not execute SQL queries.

☒ Automatic type conversion



Pydantic validates data, converts types, and supports JSON serialisation. It does not execute SQL queries.

☐ SQL query execution

Know Reason

☐ JSON serialisation



Pydantic validates data, converts types, and supports JSON serialisation. It does not execute SQL queries.

10. Which statements about GCP Dataproc are correct? [More than one option may be correct]

☒ It is used for running Hadoop and Spark clusters



Dataproc runs Hadoop/Spark clusters, supports autoscaling, and integrates with GCS. It is not fully serverless.

☒ It supports autoscaling through autoscaling policies



Dataproc runs Hadoop/Spark clusters, supports autoscaling, and integrates with GCS. It is not fully serverless.

☒ It is a serverless service

Know Reason

☒ It integrates with GCS for storage



Dataproc runs Hadoop/Spark clusters, supports autoscaling, and integrates with GCS. It is not fully serverless.

Foundations of Machine Learning

1. What is the formula for Mean Squared Error (MSE) in linear regression?

☐ $\frac{\sum(y_i - \hat{y}_i)}{n}$

Know Reason ▼

☒ $\frac{\sum(y_i - \hat{y}_i)^2}{n}$

^

MSE is the average of squared differences between actual and predicted values, that is

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

2. Which assumptions are true for linear regression? [More than one option may be correct]

☒ Linearity between predictors and response

^

Linear regression assumes linearity, constant variance, and low multicollinearity for stable coefficient estimates. Linear regression can work reasonably with categorical features but this is not an assumption that needs to be fulfilled.

☒ Homoscedasticity of residuals

^

Linear regression assumes linearity, constant variance, and low multicollinearity for stable coefficient estimates. Linear regression can work reasonably with categorical features but this is not an assumption that needs to be fulfilled.

☐ The dataset must contain only categorical features

Know Reason ▼

☒ No multicollinearity among predictors

^

Linear regression assumes linearity, constant variance, and low multicollinearity for stable coefficient estimates. Linear regression can work reasonably with categorical features but this is not an assumption that needs to be fulfilled.

3. If $z = 0$, what is the sigmoid function value $\sigma(z)$?

☐ 0

Know Reason ▼

☒ 0.5

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

For $z = 0$,

$$\sigma(0) = \frac{1}{1 + 1} = 0.5$$

4. Given a confusion matrix $TP = 50$, $FP = 10$, $FN = 5$, $TN = 35$, what is the accuracy?

☒ 0.85

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{50 + 35}{50 + 35 + 10 + 5} = \frac{85}{100} = 0.85$$

5. If a node has class distribution $[40, 60]$, what is its Gini index?

☐ 0.24

Know Reason ▼

☒ 0.48

$$\text{Gini} = 1 - \sum p_i^2 = 1 - (0.4^2 + 0.6^2) = 1 - (0.16 + 0.36) = 0.48$$

6. Which statements about decision tree hyperparameters are correct? [More than one option may be correct]

☒ max_depth controls tree depth



All listed hyperparameters are valid in scikit-learn decision trees

☐ min_samples_split controls minimum samples to split a node



All listed hyperparameters are valid in scikit-learn decision trees

☐ criterion can be 'gini' or 'entropy'



All listed hyperparameters are valid in scikit-learn decision trees

☒ min_samples_leaf sets minimum samples in a leaf node



All listed hyperparameters are valid in scikit-learn decision trees

7. What does inertia measure in K-Means?

☐ Sum of distances between cluster centroids

Know Reason ▼

☒ Sum of squared distances of samples to their closest cluster center



Inertia = Within-Cluster Sum of Squares (WCSS)

☐ Number of clusters

Know Reason ▼

☐ None of these

Know Reason ▼

8. Which statements about K-Means algorithm are correct? [More than one option may be correct]

☒ It minimises WCSS

K-Means minimises WCSS. It requires specifying K. K-Means converges to a local optimum, not global.

☒ It requires specifying number of clusters (k)

K-Means minimises WCSS. It requires specifying K. K-Means converges to a local optimum, not global.

9. Which regularisation technique adds penalty proportional to absolute value of coefficients?

☐ Ridge

Know Reason ▼

☒ Lasso

Lasso uses L1 penalty while Ridge uses L2

10. Why is regularisation used in logistic regression?

☐ To increase error

Know Reason ▼

☒ To prevent overfitting

Regularisation penalises large coefficients, reducing model complexity and leading to reduced overfitting

Deep Learning

11. A neuron has inputs $x_1 = 0.5$, $x_2 = 0.3$, weights $w_1 = 0.4$, $w_2 = 0.6$, and bias $b = 0.2$. Compute z before applying activation.

☒ 0.58

$$z = (0.5 \times 0.4) + (0.3 \times 0.6) + 0.2 = 0.2 + 0.18 + 0.2 = 0.58$$

2. Which statements about backpropagation are correct? [More than one option may be correct]

☐ It uses chain rule to compute gradients

Backpropagation applies chain rule, propagates error backward, and requires differentiable activation. Weight updates occur after gradients are computed for all layers, not just output.

☒ It updates weights after computing error at output layer only

Know Reason

☒ It propagates error backward through layers

Backpropagation applies chain rule, propagates error backward, and requires differentiable activation. Weight updates occur after gradients are computed for all layers, not just output.

☐ It requires activation function to be differentiable

Backpropagation applies chain rule, propagates error backward, and requires differentiable activation. Weight updates occur after gradients are computed for all layers, not just output.

3. Which of the following are examples of lexical processing in NLP?
[More than one option may be correct]

3/10

☒ Tokenising text into words

Lexical processing focuses on word-level operations. Tokenisation, stopword removal, and stemming directly operate on word forms. Parsing sentence structure is a syntactic task, not a lexical one.

☒ Removing stopwords from a sentence

Lexical processing focuses on word-level operations. Tokenisation, stopword removal, and stemming directly operate on word forms. Parsing sentence structure is a syntactic task, not a lexical one.

☒ Parsing sentence structure

Know Reason

☐ Converting words to their root forms (stemming)

Lexical processing focuses on word-level operations. Tokenisation, stopword removal, and stemming directly operate on word forms. Parsing sentence structure is a syntactic task, not a lexical one.

4. Why is early stopping used during the training of deep learning models?
[More than one option may be correct]

4/10

☒ To prevent overfitting by monitoring validation performance

Early stopping halts training when the model no longer improves on validation data, helping prevent overfitting and avoiding unnecessary continued learning that harms generalisation. It does not replace other regularisation techniques.

☐ To stop training when validation performance no longer improves

Early stopping halts training when the model no longer improves on validation data, helping prevent overfitting and avoiding unnecessary continued learning that harms generalisation. It does not replace other regularisation techniques.

☐ To eliminate the need for regularisation techniques

Know Reason ▼

☒ To reduce unnecessary training once the model stops generalising

Early stopping halts training when the model no longer improves on validation data, helping prevent overfitting and avoiding unnecessary continued learning that harms generalisation. It does not replace other regularisation techniques.

5. Which statement correctly distinguishes meronymy from hyponymy in semantic processing?

☒ Meronymy represents a part-whole relationship, while hyponymy represents a type-of relationship

Meronymy defines a relationship where one word denotes a part of a larger whole (e.g., "wheel" is a meronym of "car"). Hyponymy defines a hierarchical category relationship where a specific term belongs to a broader class (e.g., "sparrow" is a hyponym of "bird").

6. A language model provides the following probabilities:

$P('NLP') = 0.4$

$P('is') = 0.5$

$P('useful') = 0.3$

Assuming each word is independent, what is the probability of the sentence
"NLP is useful"?

☒ 0.06

The sentence probability is obtained by multiplying the probabilities of each word: $0.4 \times 0.5 \times 0.3 = 0.06$. So, the probability of the sentence is 0.06.

7. During training of a neural network, you notice that the training accuracy continues to increase, but the validation accuracy starts to decrease. What does this behaviour most strongly indicate?

☐ The model is underfitting

Know Reason ▼

☒ The model is overfitting

^

When a model performs well on training data but poorly on unseen validation data, it indicates that the model has memorised the training patterns instead of learning generalisable features. This is a classic sign of overfitting.

8. Which NLP task assigns grammatical categories such as noun, verb, and adjective to each word in a sentence?

☐ Tokenisation

Know Reason ▼

☒ POS tagging

^

POS tagging identifies the grammatical role of each word, enabling deeper syntactic analysis such as parsing and grammatical agreement checking.

9. A neural network is designed for a multiclass classification problem with 3 classes using Softmax at the output layer. What will be the dimension of the output layer vector?

☐ (1, 1)

Know Reason ▼

☒ (1, 3)

^

In multiclass classification with 3 target classes, the output layer must produce one output value per class. Softmax converts these into a probability distribution across all classes. Therefore, the output vector must have 3 elements, giving an output dimension of (1, 3).

10. Why is the Bag-of-Words model considered limited for semantic understanding?

☐ It captures word order and context precisely

Know Reason ▼

☐ It represents text using semantic vector spaces

Know Reason ▼

☒ It treats words as independent frequency features

^

Bag-of-Words ignores semantic relationships and word order, representing text purely through frequency counts, which limits meaningful interpretation.

```
import ast
```

```
# Input is taken for you
```

```
stringslist = ast.literal_eval(input())
```

```
def transformstrings(strings):
```

```
    if not strings:
```

```
        return ['EMPTY']
```

```
    # Filter strings with length >= 3, reverse and uppercase them
```

```
    transformed = [
```

```
        (len(s), s[::-1].upper())
```

```
        for s in strings
```

```
        if len(s) >= 3
```

```
    ]
```

```
    # Sort by original length in descending order
```

```
    transformed.sort(key=lambda x: x[0], reverse=True)
```

```
    # Return only the transformed strings
```

```
    return [t[1] for t in transformed]
```

```
print(transformstrings(stringslist))
```

```
# Do not edit this data processing section
```

```
import pandas as pd
```

```
df = pd.read_csv('https://d3ejq4mxgimsmf.cloudfront.net/sales_data_1-8dc2fb312fd14ba590754784a97dd3ea.csv')
```

```
# Code here
```

```
temp = df.describe().T
```

```
temp['lower'] = temp['25%'] - 1.5 * (temp['75%'] - temp['25%'])
```

```
temp['upper'] = temp['75%'] + 1.5 * (temp['75%'] - temp['25%'])
```

```
temp = temp[['lower', 'upper']]
```

```
i = -1
```

```
c = [0, 0, 0]
```

```
for col in ['total_sales', 'customer_count', 'average_transaction_value']:
```

```
    i += 1
```

```
    c[i] = (df[col] > temp.loc[col, 'lower']) & (df[col] < temp.loc[col, 'upper'])
```

```
df = df.loc[c[0] & c[1] & c[2]]
```

```
# Do not edit this input-output processing section
```

```
df = df.reset_index(drop = True)
```

```
z = input().split(',')
```

```
x = int(z[0].strip())
```

```
y = z[1].strip()
```

```
print(df.loc[x, y])
```

C1 Advanced Mathematics and Programming for ML

2. Now, the bank has the following data:

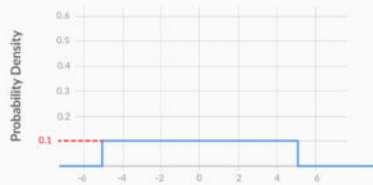
Defaulter/Non-Defaulter	Probability	X (Money Lost by Bank)
Defaulter	0.007	₹9.2 lakh
Non-Defaulter	0.993	0

What would be the expected loss (expected value of the money lost by the bank)?

EDMLDS - CIMIS1 - Practice Questions - Q2

✓ ₹6,440

Correct! The random variable X has two possible values: ₹9.2 lakh for when the customer defaults and 0 for when the customer does not. So, the expected value = ₹9.2 lakh * $P(X = 9.2 \text{ lakh})$ + 0 * $P(X=0)$ = ₹9,20,000 * (0.007) + 0 * (0.993) = ₹6,440.



The graph you see above represents the PDF of a uniformly distributed random variable X . As you can see, the probability density is equal for all the possible values of X (-5 to +5).

What is the probability of the random variable X lying between -1.5 and +2.5, i.e., $P(-1.5 < X < 2.5)$?

EDMLDS - CIMIS3 - Graded Questions - Q1

0.1

0.4

Your Answer

Correct Answer



Correct! The probability of the variable lying between -1.5 and 2.5 would be equal to the area under the PDF between $X = -1.5$ and $X = 2.5$. This would be equal to the area of a rectangle with a breadth of 0.1 and a length of 4, i.e., $2.5 - (-1.5)$. By multiplying the two, you get the area of the rectangle, which is 0.4.

2. The **normal distribution**, aka the **Gaussian distribution**, was discovered by **Carl Friedrich Gauss** in 1809. Gauss was trying to create a probability distribution for **astronomical errors**. Astronomical errors are the errors that were made by astronomers while observing phenomena such as distances in space. ✔ Single Choice

For example, Gauss found that an astronomer trying to estimate the distance between Earth and Uranus always makes an error.

This **error** is **normally distributed**, with $\mu = 0$ km and $\sigma = 1,000$ km.

Based on the information above, what is the probability of the astronomer overestimating the distance by 2,330 km or more?

(You can refer an external Z-table for help here)

EDMLDS - CIMIS3 - Graded Questions - Q2

1%

Your Answer

Correct Answer



Let's define X as the astronomical error, which is normally distributed with mean 0 km and standard deviation 1,000 km. Now, you have to find the probability that $X > 2330$, i.e. $P(X > 2330)$. Converting this to Z, it becomes $P(Z > 2.33)$. Since $P(Z < -2.33) + P(Z > 2.33) = 1$, $P(Z > 2.33) = 1 - P(Z < 2.33) = 1 - 0.9901 = 0.0099$ or 0.99%, which is approximately 1%.

3. The **normal distribution**, aka the **Gaussian distribution**, was discovered by **Carl Friedrich Gauss** in 1809. Gauss was trying to create a probability distribution for **astronomical errors**. Astronomical errors are the errors that were made by astronomers while observing phenomena such as distances in space. ✔ Single Choice

For example, Gauss found that an astronomer trying to estimate the distance between Earth and Uranus always makes an error.

This **error** is **normally distributed**, with $\mu = 0$ km and $\sigma = 1,000$ km.

Hence, what is the probability that the astronomer under- or over-estimates the distance by less than 500 km?

EDMLDS - CIMIS3 - Graded Questions - Q3

30.85%



69.15%



38.30%

Your Answer

Correct Answer



Let's define X as the astronomical error, which is normally distributed with mean 0 km and standard deviation 1,000 km. Now, you have to find the probability that $-500 < X < 500$, i.e. $P(-500 < X < 500)$. Converting this to Z, it becomes $P(-0.5 < Z < 0.5) = P(Z < 0.5) - P(Z < -0.5) = 0.6915 - 0.3085 = 0.3830$, or 38.30%.

C3 Cloud Computing and Big Data Fundamentals

C3 M1 ETL with PySpark

1. What does ETL stand for in data applications?

1/10

☐ Extract, transfer, load

Know Reason ▼

☐ Execute, transform, load

Know Reason ▼

☒ Extract, transform, load

^

C is correct: ETL stands for Extract, Transform, Load — the standard workflow for preparing data.

2. Which of the following is an example of a structured data source?

2/10

☐ Tweets

Know Reason ▼

☒ CSV file

^

B is Correct: CSV files are structured, with rows and columns.

☐ Scanned PDFs

Know Reason ▼

3. What is involved in ETL transformation?

☐ Copying unprocessed data

Know Reason ▼

☐ Encrypting data for protection

Know Reason ▼

☒ Preparing and cleaning data

^

C is correct: Transformation involves data enrichment, standardisation and cleaning.

☐ Data extraction from APIs

Know Reason ▼

4. What is incremental extraction?

☐ Extracting every record from the source

Know Reason ▼

☒ Extracting only updated or new information

^

B is Correct: Only new or updated records are pulled by incremental extraction.

☐ Extracting information according to time zone

Know Reason ▼

☐ Starting with the biggest partitions

Know Reason ▼

5. Which stage of the transformation guarantees that the dates are formatted consistently?

☐ Cleaning

Know Reason ▼

☐ Deriving

Know Reason ▼

☒ Standardising

^

C is correct: Standardising is the transformation step that ensures data (such as dates, names, formats, etc.) follows a consistent structure or format across the dataset.

☐ Encrypting

Know Reason ▼

6. What is batch loading?

☐ Data can be loaded in real time

Know Reason ▼

☐ Loading all data every second

Know Reason ▼

☒ Loading data at predetermined intervals

^

C is correct: Batch loading occurs at predetermined intervals.

☐ Loading one record at a time

Know Reason ▼

7. Which file format is used by default in the majority of ETL practice data sets?

Note: More than one option may be correct

☒ XML

Know Reason ▼

☒ JSON

^

B is Correct: JSON is also used, but it used less frequently than CSV as it is much simpler to use.

☒ AVRO

Know Reason ▼

☒ CSV

^

8. Why is PySpark appropriate for large scale ETL?

☒ It supports both distributed and in-memory computing.

^

A is Correct: PySpark uses memory and clusters to process large data sets.

☐ It utilises Java.

Know Reason ▼

☐ It is GPU-based.

Know Reason ▼

☐ It uses less RAM.

Know Reason ▼

9. In PySpark, which Spark component carries out your ETL logic?

☐ HiveServer

Know Reason ▼

☒ SparkSession



B is Correct: PySpark applications start with SparkSession.

☐ The DataNode

Know Reason ▼

☐ KafkaBroker

Know Reason ▼

10. Which PySpark command generates a temporary SQL view?

10/10

Note: More than one option may be correct

☐ createOrReplaceTempView()



A is Correct: This is the correct PySpark method to register a DataFrame as a SQL table.

☒ createTempView()



B is Correct: This is deprecated in more recent versions of PySpark.

☐ registerTempTable()

Know Reason ▼

☒ sparkSQLView()

Know Reason ▼

1.

Which PySpark feature ensures resilience during computation?

✓ Single Choice

DAG

Lazy evaluation

RDD

Your Answer

Correct Answer



Pandas UDF

2. Which operation triggers execution in PySpark?

✓ Single Choice

filter()

withColumn()

show()

Your Answer

Correct Answer



select()

3. Which file format is columnar and optimised for PySpark?

✓ Single Choice

JSON

XML

CSV

Parquet

Your Answer

Correct Answer



4. Which benefit does PySpark's in-memory processing offer?

✓ Single Choice

Quicker computation

Your Answer

Correct Answer



Less RAM usage

Cheaper storage

Simpler syntax

5. Why do most users prefer PySpark DataFrames over RDDs?

✓ Single Choice

They are SQL-optimised and simpler to write.

Your Answer

Correct Answer



They use Java.

They permit UDFs.

They support Pandas commands.

6. Which is a recommended use case for UDFs in PySpark?

✓ Single Choice

Basic filtering

Standard joins

Complex text transformation

Your Answer

Correct Answer



Type casting

7. What PySpark function reads a CSV file into a DataFrame?

✓ Single Choice

`readFile("file.csv")`

`read.csv("file.csv")`

Your Answer

Correct Answer



`csv.load("file.csv")`

`read_text("file.csv")`

8. What function does Spark's DAG serve?

✓ Single Choice

Handle file formats

Make SQL syntax better

Show memory

Monitor and improve transformations

Your Answer

Correct Answer



9. Which PySpark method allows filtering a DataFrame?

✓ Single Choice

groupBy()

select()

show()

filter()

Your Answer

Correct Answer



10. Why should you refrain from using UDFs excessively in PySpark?

✓ Single Choice

They cause the session to crash.

They bypass the DAG.

They impede optimisation and slow down performance.

Your Answer

Correct Answer



They only work with JSON.

C3 M2 ETL on Cloud

1. A startup uses cloud-based ETL but wants complete control over the OS, middleware and runtime while managing costs. Which service model best fits their needs?

☐	SaaS	Know Reason ▼
☐	PaaS	Know Reason ▼
☒	IaaS	^
NA		
☐	FaaS	Know Reason ▼

2. A company wants to use cloud infrastructure for a few workloads while keeping sensitive workloads on-premises. Which deployment model should they adopt?

☐	Public Cloud	Know Reason ▼
☒	Hybrid Cloud	^
NA		
☐	Community Cloud	Know Reason ▼
☐	Multi-Cloud	Know Reason ▼

3. Which of the following is a consequence of vendor lock-in in cloud computing?

☐ Faster data migration across providers

Know Reason ▼

☐ Easy switching between cloud providers

Know Reason ▼

☒ Dependency on a specific provider's tools and APIs

^

NA

☐ Increased fault tolerance

Know Reason ▼

4. In the shared responsibility model, who is responsible for data classification and access controls in a cloud-hosted SaaS application?

☐ Cloud provider only

Know Reason ▼

☒ End user only

^

NA

☒ Both provider and user

Know Reason ▼

☐ Cloud provider's DevOps team

Know Reason ▼

5. An application needs to scale automatically based on traffic. Which combination of services would be best suited in a cloud-native environment?

☒ Auto-scaling + Object Storage + Serverless functions ^

NA

☐ Virtual machines + CDN + Multi-region backup Know Reason v

☐ Static web hosting + VPN + Load balancer Know Reason v

☐ Bare-metal server + Manual monitoring Know Reason v

6. Which of the following will most likely impact performance negatively in a cloud-based ETL job if not optimised?

☐ Number of API calls Know Reason v

☐ Lack of encryption Know Reason v

☐ Improper use of availability zones Know Reason v

☒ Unpartitioned data and large shuffles ^

NA

7. What will be the output of the following PySpark operation?
`df.groupBy("category").agg({"amount": "sum", "amount": "avg"})`



Returns both sum and avg of "amount" by "category"

Know Reason ▼



Throws an error due to duplicate key in aggregation

Know Reason ▼



Only calculates sum and ignores avg

Know Reason ▼



Only calculates avg and ignores sum



Correct. In Python, when a dictionary contains duplicate keys, the last key-value pair overwrites the earlier ones. In this aggregation, "amount": "avg" overwrites "amount": "sum", so only the average is calculated, and the sum is ignored.

8. Which of the following transformations in PySpark can cause a full shuffle of the data across partitions?



`select()`

Know Reason ▼



`filter()`

Know Reason ▼



`repartition()`



NA



`withColumn()`

Know Reason ▼

9. You are dealing with a skewed dataset in PySpark where 80% of rows belong to a single key during `groupBy()`. Which approach is most effective to reduce skew?

☐ Use `cache()` before `groupBy`

Know Reason ▼

☐ Increase executor memory

Know Reason ▼

☒ Apply salting to the key before grouping

^

NA

☐ Use broadcast join instead

Know Reason ▼

10. Which of the following most accurately describes how `persist(StorageLevel.MEMORY_AND_DISK)` works in PySpark?

☐ Caches the RDD in memory only; throws an error if memory is insufficient

Know Reason ▼

☐ Writes RDD to disk only

Know Reason ▼

☒ Stores RDD in memory and spills to disk if memory is insufficient

^

NA

☐ Stores RDD in memory and recomputes partitions when needed

Know Reason ▼

1. Which cloud model is best suited for an organisation that must comply with strict data residency laws and still wants flexibility in bursting workloads into public cloud environments?

✓ Single Choice

Public cloud

Private cloud

Hybrid cloud

Your Answer

Correct Answer



Multi-cloud

2. Which of the following best explains why using `collect()` on large DataFrames is discouraged in PySpark?

✓ Single Choice

It loads only null values to the driver.

It blocks executor threads but not the driver.

It loads all partitions into driver memory and can cause OOM.

Your Answer

Correct Answer



It triggers a recomputation of RDD lineage.

3. Which of the following best explains why cloud-native ETL tools are preferred over traditional ETL tools in modern pipelines?

✓ Single Choice

They rely solely on manual configurations.

They process data only in batches.

They leverage distributed and scalable architecture.

Your Answer

Correct Answer



They require dedicated hardware setups.

4. In cloud service models, which layer hides the most complexity from end-users while limiting customisability?

✓ Single Choice

IaaS

SaaS

Your Answer

Correct Answer



PaaS

FaaS

5. Which characteristic of cloud computing is most important when performing **time-bound analytics** over large datasets?

✔ Single Choice

Pay-as-you-go billing

Resource pooling

Rapid elasticity

Your Answer

Correct Answer



Vendor independence

6. Which Spark capability enables handling batch, streaming, SQL, and ML workloads in a single platform?

✔ Single Choice

Auto-scaling

Unified analytics engine

Your Answer

Correct Answer



Persistent RDDs

Spark Explorer Dashboard

7. Which scenario best illustrates a false sense of high availability in a multi-region architecture?

✓ Single Choice

Using multiple load balancers for different services

Hosting failover resources in a disabled region

Your Answer

Correct Answer



Replicating all data to another region using active-active mode

Placing an autoscaler in only one region

8. In PySpark, which of the following will result in narrow dependency between RDDs/DataFrames?

✓ Single Choice

repartition(10)

join()

groupBy()

map()

Your Answer

Correct Answer



9. Given a large DataFrame `df`, what is the most memory-efficient way to perform a join with a very small lookup DataFrame `df_lookup`?

✔ Single Choice

Use `df.join(df_lookup)` directly.

Use `broadcast(df_lookup)` before join.

Your Answer

Correct Answer



Cache both DataFrames.

Perform a `crossJoin` with filters.

10. In a PySpark job, a user observes that after applying `persist(StorageLevel.DISK_ONLY)`, subsequent actions still take a long time. What is a likely explanation?

✔ Single Choice

The cluster has insufficient memory.

Disk persistence only avoids recomputation, but access speed is slow.

Your Answer

Correct Answer



The `persist` operation didn't trigger due to lazy evaluation.

`Persist` only works on broadcast joins.

C3 M3 Containerised Data Science Applications

1. Which of the following is a key advantage of FastAPI for ETL APIs?

☐ Native data visualisation dashboards Know Reason ▼

☐ Built-in machine learning model training Know Reason ▼

☐ Preconfigured cloud deployments Know Reason ▼

☒ Automatic Swagger documentation from type hints ^

Correct. FastAPI auto-generates interactive docs using type hints.

2. What role does Pydantic play in API development?

☐ It handles caching for ETL pipelines. Know Reason ▼

☒ It validates input and output schemas. ^

Correct. Pydantic enforces clean, predictable schemas.

☐ It monitors cluster performance. Know Reason ▼

☐ It manages async background jobs. Know Reason ▼

3. Why is async def important when handling large-scale data?



It guarantees faster computation of Spark jobs.

Know Reason ▼



It prevents server blocking during I/O operations.



Correct. Async keeps servers responsive during long I/O waits.



It makes APIs automatically more secure.

Know Reason ▼



It replaces the need for cloud storage.

Know Reason ▼

4. Why do we use Docker in ETL workflows?



To package apps with dependencies for consistent deployment



Correct. Docker ensures consistency across environments.



To automatically scale Spark clusters

Know Reason ▼



To generate API documentation

Know Reason ▼



To manage relational databases

Know Reason ▼

5. What does a docker-compose.yml file do?



Creates new Python virtual environments

Know Reason ▼



Uploads images directly to Docker Hub

Know Reason ▼



Automates Spark ETL code generation

Know Reason ▼



Defines and manages multi-container applications



Correct. Compose orchestrates multiple services together.

6. Which of the following is an advantage of containerisation over running apps directly on VMs?

☐ Containers guarantee infinite storage.

Know Reason ▼

☐ Containers replace the need for APIs.

Know Reason ▼

☐ Containers eliminate the need for monitoring.

Know Reason ▼

☒ Containers use less overhead and start faster.

^

Correct. Containers are lightweight compared with VMs.

7. What is the main purpose of Databricks in ETL workflows?

☒ Replacing all cloud storage providers

Know Reason ▼

☐ Acting as a replacement for Docker

Know Reason ▼

☐ Hosting static HTML dashboards

Know Reason ▼

☒ Providing collaborative notebooks and scalable Spark compute

^

Correct. Databricks unites notebooks, Spark clusters and workflows.

8. What does a Delta Lake add to raw data lakes?

☐ Streaming video support

Know Reason ▼

☐ Built-in machine learning models

Know Reason ▼

☐ Faster UI rendering

Know Reason ▼

☒ Schema enforcement and ACID transactions

^

Correct. Delta Lake ensures reliable, consistent data.

9. What is the role of Databricks Workflows (Jobs)?

☐ To train neural networks

Know Reason ▼

☐ To replace storage connectors

Know Reason ▼

☐ To provide visualisation dashboards

Know Reason ▼

☒ To execute and schedule notebooks automatically

^

Correct. Workflows automate ETL notebook execution.

10. Which of the following is the core advantage of Streamlit?

☐ It provides Spark cluster orchestration.

Know Reason ▼

☐ It replaces databases with in-memory caching.

Know Reason ▼

☐ It auto-scales APIs to the cloud.

Know Reason ▼

☒ It turns Python scripts into interactive dashboards with minimal code.

^

Correct. Streamlit makes dashboards from Python easily.

11. Which Streamlit component allows file uploads?

☐ `st.table`

Know Reason ▼

☐ `st.slider`

Know Reason ▼

☒ `st.file_uploader`

^

Correct. `st.file_uploader` is used for file uploads

☐ `st.button`

Know Reason ▼

12. What is the purpose of `.streamlit/secrets.toml`?

☐ To define container orchestration

Know Reason ▼

☐ To deploy Streamlit apps on AWS

Know Reason ▼

☒ To securely store credentials and keys

^

Correct. `Secrets.toml` stores sensitive configurations securely.

☐ To replace pandas configurations

Know Reason ▼

3. Which cloud option makes it easiest to deploy Streamlit apps?

☒ Streamlit Cloud



Correct. Streamlit Cloud is the simplest choice.

☐ AWS EC2

Know Reason ▼

☐ Azure VM

Know Reason ▼

☐ Google App Engine

Know Reason ▼

4. What is the purpose of a job-trigger endpoint such as /run-etl?

☐ To secure Flask apps

Know Reason ▼

☐ To monitor VM health

Know Reason ▼

☒ To manually start ETL pipelines via API calls



Correct. /run-etl triggers ETL jobs manually.

☐ To schedule jobs automatically without APIs

Know Reason ▼

5. What is the role of `/get-status` in an ETL API?

☒ It checks job progress and completion. ^

Correct. Status APIs return progress or completion details.

☐ It secures the Flask app with tokens.

Know Reason v

☐ It uploads data to databases.

Know Reason v

☐ It configures Docker Compose services.

Know Reason v

6. What is a webhook in ETL terms?

☒ An event-driven callback that pushes data to your API ^

Correct. Webhooks deliver real-time data when events occur.

☐ A Docker command for container orchestration

Know Reason v

☐ A monitoring dashboard feature

Know Reason v

☐ A scheduler for batch pipelines

Know Reason v

7. Which component acts as the message broker in a Flask + Celery stack?



Celery

Know Reason ▼



SQLite

Know Reason ▼



Docker

Know Reason ▼



Redis



Correct. Redis brokers tasks between Flask and Celery.

8. What is the role of Celery in ETL services?



To monitor Flask security

Know Reason ▼



To execute background tasks asynchronously



Correct. Celery runs long ETL tasks in the background.



To configure VM scaling

Know Reason ▼



To act as the database engine

Know Reason ▼

9. Why is Docker Compose useful for deploying Flask ETL services?



It configures pandas transformations.

Know Reason ▼



It orchestrates multiple containers such as Flask, Redis and Celery.



Correct. Compose manages multiple services together.



It auto-generates Swagger docs.

Know Reason ▼



It replaces webhooks with schedulers.

Know Reason ▼

10. What is the overall benefit of combining FastAPI, Docker, Databricks, Streamlit and Flask?

☐ To replace SQL with Python completely

Know Reason 

☒ To build flexible, scalable and cloud-ready ETL workflows 

Correct. Together, these tools form a modern ETL toolkit.

☐ To create a monolithic ETL system

Know Reason 

☐ To eliminate the need for DevOps practices

Know Reason 

1. Which of the following best explains the difference between serverful and serverless architectures?

  Single Choice

Serverful architecture means you provision and manage infrastructure yourself, whereas serverless architecture abstracts this away.

Your Answer

Correct Answer



Serverful architecture is cheaper than serverless architecture.



Serverless architecture means you cannot run Python applications.



Serverful architecture is only used for machine learning workloads.



2. In FastAPI, what does the command `uvicorn main:app --reload` do?

  Single Choice

It deploys the app to the cloud.



It starts the FastAPI server with auto-reload for development.

Your Answer

Correct Answer



It compiles Python code into bytecode.



It generates database migrations.



3. Why are type hints useful in FastAPI?

✓ Single Choice

They allow FastAPI to validate inputs and generate interactive documentation.

Your Answer Correct Answer ✓

They make Python code execute faster.

▼

They automatically convert Python code into Java.

▼

They replace the need for testing.

▼

4. What is a common trade-off associated with serverless functions?

✓ Single Choice

Infinite memory

▼

Vendor lock-in and cold starts

Your Answer Correct Answer ✓

Built-in database support

▼

Guaranteed zero cost

▼

5. Which scenario is best suited for a serverless ETL function?

✓ Single Choice

A banking system requiring full control of infrastructure



An unpredictable data ingestion pipeline with inconsistent workloads

Your Answer

Correct Answer



A traditional ERP system with stable workloads



A large-scale Hadoop cluster



6. In Docker, what is the purpose of an image?

✓ Single Choice

It acts as a live running container.



It stores the definition of an application and its dependencies.

Your Answer

Correct Answer



It is a graphical representation of a container.



It configures Kubernetes clusters.



7. What does the docker run command do?

✓ Single Choice

It uploads an image to Docker Hub.



It compiles a Python application.



It creates and starts a container from an image.

Your Answer

Correct Answer



It configures environment variables globally.



8. Which file is commonly used to define dependencies and instructions to build a Docker image?

✓ Single Choice

docker-compose.yml



requirements.txt



Dockerfile

Your Answer

Correct Answer



.env



9. Why might a team prefer Docker Compose over running single Docker commands?

✓ Single Choice

It allows defining multiple containers and their interactions in one place.

Your Answer Correct Answer ✓

It automatically deploys apps to Kubernetes.

▼

It eliminates the need for Dockerfiles.

▼

It is used only for building machine learning models.

▼

10. In a containerised ETL setup, why might you use Redis with Docker?

✓ Single Choice

To handle the schema validation of API inputs

▼

To store files permanently

▼

To act as a message broker between services

Your Answer Correct Answer ✓

To generate Swagger docs

▼

C3 M4 Application Deployment

1. What does *serverless architecture* primarily mean?

☐ No servers are used at all.

Know Reason ▼

✓ ☒ Developers no longer need to manage servers directly.

^

Correct! Cloud providers handle provisioning, scaling, and maintenance.

☐ Applications cannot scale automatically.

Know Reason ▼

☐ Applications always run continuously.

Know Reason ▼

2. Which of the following is a key benefit of serverless deployments?



Automatic scalability.



Correct!! Serverless adjusts to demand without manual intervention.



Cost efficiency, paying only when code runs.



Correct!! You're charged per execution, not for idle time.



Full manual control of infrastructure.

Know Reason



Complex server configuration is always required.

Know Reason

3. Which service is Amazon's main offering for serverless computing?



AWS Lambda.



Correct!! Lambda is AWS's core event-driven serverless service.



Amazon Redshift.

Know Reason



Amazon EC2.

Know Reason



Amazon CloudWatch.

Know Reason

4. When comparing serverless offerings, which feature is most associated with Google Cloud Functions?



Seamless developer tools integration.

Know Reason



Strong ecosystem integration.

Know Reason



Global scalability with transparent pricing.



Correct!! GCF is designed for simplicity, real-time use cases, and mobile-first apps.

5. What is the purpose of *dependency management* in serverless deployments?

☐ To configure runtime environments securely.

Know Reason ▼

☒ To package libraries and tools with your code. ^

Correct! Dependencies need to be bundled for serverless execution.

☐ To eliminate the need for environment variables.

Know Reason ▼

☐ To bypass cloud security restrictions.

Know Reason ▼

6. Which cloud monitoring tool allows distributed tracing for serverless applications?

☐ AWS CloudWatch Metrics.

Know Reason ▼

☒ Azure Application Insights. ^

Correct! It supports distributed tracing across multiple services.

☒ Google Cloud Logging.

Know Reason ▼

7. In GCP, which service is commonly used for large-scale ETL processing in a serverless pipeline?

☐ Cloud Functions

Know Reason ▼

☒ Dataproc ^

Correct! Dataproc executes PySpark ETL jobs at scale.

☒ BigQuery

Know Reason ▼

☐ Cloud Storage

Know Reason ▼

8. Which of the following statements about *logging in serverless environments* is true?

☐ Logs are stored locally on the developer's machine

Know Reason ▼

☒ AWS Lambda logs are automatically sent to CloudWatch

^

Correct! This provides centralised monitoring and error tracking.

☐ Azure Functions cannot produce logs

Know Reason ▼

☐ Google Cloud Functions do not support log filtering

Know Reason ▼

9. Which of these is an advantage of using Git LFS in data science projects?

☒ It avoids storing large files directly in Git repositories

^

Correct! Git LFS replaces them with lightweight pointers.

☐ It eliminates the need for version control altogether

Know Reason ▼

☐ It prevents teammates from collaborating.

Know Reason ▼

☐ It automatically merges all conflicts without review

Know Reason ▼

10. Which branching strategy promotes frequent commits to the main branch and supports continuous integration?

☐ Git Flow

Know Reason ▼

☒ Trunk-Based Development

^

Correct! Developers commit regularly to the main branch.

☐ Forking workflow

Know Reason ▼

1. Where can the processed outputs of Dataproc ETL jobs be stored?
(More than one option may be correct.)

✓ ☒ Multiple Choice

BigQuery for structured querying

Your Answer

Correct Answer



Google Cloud Storage in CSV or Parquet formats

Your Answer

Correct Answer



Directly in AWS Redshift

Only on local developer machines

2. Which AWS service is used to run PySpark ETL jobs in a serverless manner?

✓ ☐ Single Choice

AWS Lambda

EMR Serverless

Your Answer

Correct Answer



S3

API Gateway

3. In Azure, which feature enables orchestration of multi-step workflows in serverless environments?

✓ ☐ Single Choice

Durable Functions

Your Answer

Correct Answer



Azure Functions

Azure Data Lake

Azure Blob Storage

4. A data engineering team wants to build a workflow on GCP that automatically processes uploaded CSV files, runs transformations in PySpark, and stores results in BigQuery. Which combination of services best suits this use case?

✓ ☐ Single Choice

Cloud Storage, App Engine, Compute Engine

Pub/Sub, Firebase, BigQuery

Dataproc only

Cloud Storage, Cloud Functions, Dataproc, BigQuery

Your Answer

Correct Answer



3. In Azure, which feature enables orchestration of multi-step workflows in serverless environments?

✓ Single Choice

Durable Functions

Your Answer

Correct Answer



Azure Functions

Azure Data Lake

Azure Blob Storage

4. A data engineering team wants to build a workflow on GCP that automatically processes uploaded CSV files, runs transformations in PySpark, and stores results in BigQuery. Which combination of services best suits this use case?

✓ Single Choice

Cloud Storage, App Engine, Compute Engine

Pub/Sub, Firebase, BigQuery

Dataproc only

Cloud Storage, Cloud Functions, Dataproc, BigQuery

Your Answer

Correct Answer



5. Your team deployed an AWS Lambda function triggered by an S3 event. The function must submit PySpark jobs for large datasets. Which AWS service should it connect to for this?

✓ Single Choice

Amazon EC2

EMR Serverless

Your Answer

Correct Answer



Amazon Redshift

CloudFormation

6. A team is building a serverless API on AWS that should handle web requests and trigger Spark jobs. Which services will be most useful here?

✓ Single Choice

API Gateway, Lambda, EMR Serverless

Your Answer

Correct Answer



S3, CloudFront, Route 53

CloudTrail, IAM, KMS

DynamoDB, Redshift

7. Which of the following best describes the role of Azure Durable Functions?

✓ Single Choice

They provide persistent storage for large datasets

They orchestrate multi-step workflows with state management

Your Answer

Correct Answer



They handle only simple file transformations

They act as an API gateway for Azure.

8. A developer uploads a large dataset to Azure Blob Storage. Immediately, an ETL process starts automatically without manual intervention. Which Azure component enabled this automation?

✓ Single Choice

Blob-triggered Azure Functions

Your Answer

Correct Answer



Azure DevOps Pipelines

Azure Synapse

Virtual Machines

9. Your team wants to track changes in Jupyter Notebooks without cluttering Git history with outputs. Which tool or practice should they adopt?  Single Choice

Use nbdtm for diffing and merging notebooks

Your Answer

Correct Answer



Disable Git version control for notebooks

Always store large outputs directly in Git.

Avoid using Jupyter Notebooks in projects.

10. A company is deploying a FastAPI application at scale and wants to manage multiple containers across frontend, backend, and database layers. Which Kubernetes components will help maintain reliability and connectivity?  Single Choice

IAM, CloudTrail, CloudWatch

API Gateway, Lambda, DynamoDB

Buckets, Blobs, Queues

Pods, Deployments, Services

Your Answer

Correct Answer



Chat GPT Questions

SECTION A – MCQs (60 Questions)

C1 – Mathematics (1–15)

1. If $P(A)=0.4$, $P(B)=0.5$, $P(A \cap B)=0.2$, then $P(A|B)=?$
0.4
 2. Which distribution models number of successes in fixed trials?
Binomial
 3. Expected value of fair dice?
 $E(X)=\sum x \cdot P(x)$
 $E(X)=(1+2+3+4+5+6)/6 = 3.5$
 4. Standard deviation measures: **Spread**
 5. Z-score formula is: $(X-\mu)/\sigma$
 6. PDF integrates to: **1**
 7. Eigenvalues are solutions of: $\det(A-\lambda I)=0$
 8. Gradient is: **vector of partial derivatives**
 9. Chain rule is used in: **Back Propagation**
 10. If covariance > 0 , relation is: **Positive Relation**
 11. Binomial assumes: **Independent Trials**
 12. Central Limit Theorem applies to: **Sample Means**
 13. Determinant of 2x2 matrix formula: **ad-bc**
 14. Logistic function output range: **(0,1)**
 15. Variance equals: $E[(X-\mu)^2]$
-

C1 – SQL & MongoDB (16–30)

16. Which clause filters aggregated result? **Having**
17. Which join keeps all left records? **Left Join**

- 18. COUNT(*) counts: **All rows**
 - 19. Normalization reduces: **redundancy**
 - 20. Primary key must be: **unique & not null**
 - 21. MongoDB stores data as: **documents**
 - 22. find() returns: **cursor**
 - 23. Window function example: **ROW_NUMBER**
 - 24. GROUP BY used with: **aggregate**
 - 25. ACID ensures: **consistency**
 - 26. LEFT JOIN returns null when: **no match**
 - 27. Subquery inside WHERE is: **nested query**
 - 28. Index improves: **query speed**
 - 29. ER diagram represents: **Entities and Relationships**
 - 30. Aggregation pipeline used for: **data processing**
-

C1 – Python & Libraries (31–40)

- 31. Lambda creates: **Anonymous Function**
 - 32. map() returns: **Iterator**
 - 33. Pandas groupby used for: **aggregation**
 - 34. NumPy faster because: **Vectorized operations**
 - 35. OOP promotes: **modularity**
 - 36. **init** is: **Constructor**
 - 37. Seaborn built on: **matplotlib**
 - 38. List comprehension creates: **new List**
 - 39. reduce() needs: **function**
 - 40. Inheritance allows: **code reuse**
-

C2 – Statistics & EDA (41–55)

- 41. Null hypothesis assumes: **no difference**

- 42. Type I error: **false positive**
 - 43. p-value represents: **Probability under H0**
 - 44. Z-test used when: **σ known**
 - 45. t-test used when: **σ unknown**
 - 46. Correlation range: **-1 to 1**
 - 47. IQR method detects: **outliers**
 - 48. Skewness >0 means: **right skew**
 - 49. CLT sample size approx: **30**
 - 50. Two-tailed test checks: **both sides**
 - 51. Confidence interval includes: **true mean**
 - 52. Power of test relates to: **detect effect**
 - 53. Standard error decreases when: **n increases**
 - 54. Outliers affect: **mean**
 - 55. Variance always: **non-negative**
-

C3–C5 – Cloud, ML, DL (56–60)

- 56. ETL stands for: **Extract Transform Load**
 - 57. Docker provides: **Containerization**
 - 58. Kubernetes manages: **Containers**
 - 59. Bagging reduces: **variance**
 - 60. Long Short-Term Memory (LSTM) solves: **vanishing gradient**
-

SECTION B – SHORT ANSWERS (50 Questions)

Answer in 3–5 lines.

Below are **concise 3–5 line exam-ready answers** for Q61–100.

61. State Bayes Theorem

Bayes Theorem calculates conditional probability.

$$P(A|B) = P(B|A)P(A) / P(B)$$

It updates prior belief using new evidence.

Used widely in ML and medical diagnosis.

62. Define Expected Value

Expected value is the long-run average outcome of a random variable.

$$E(X) = \sum xP(x)$$

For continuous variables, integration is used.

It represents the mean of a probability distribution.

63. Difference PDF vs CDF

PDF gives probability density at a point (continuous case).

CDF gives cumulative probability up to a value.

CDF is integral of PDF.

CDF ranges from 0 to 1.

64. Explain Gradient Descent

Gradient descent is an optimization algorithm.

It minimizes a cost function by updating parameters.

$$\theta = \theta - \alpha \frac{\partial J}{\partial \theta}$$

Used in regression and neural networks.

65. What is Eigenvector?

An eigenvector is a non-zero vector that changes only in scale when multiplied by a matrix.

$$Av = \lambda v$$

λ is the eigenvalue.

Used in PCA and dimensionality reduction.

66. Define Normalization in DB

Normalization organizes database tables to reduce redundancy.

It ensures data integrity.

Involves Normal Forms (1NF, 2NF, 3NF).

Prevents update anomalies.

67. Explain ACID

ACID ensures reliable DB transactions.

A – Atomicity

C – Consistency

I – Isolation

D – Durability

68. INNER vs LEFT JOIN

INNER JOIN returns matching rows from both tables.

LEFT JOIN returns all rows from left table and matching from right.

Unmatched rows in right become NULL.

69. What is NoSQL?

NoSQL is a non-relational database system.

Stores data as documents, key-value, graph, etc.

Flexible schema.

Example: MongoDB.

70. Explain Aggregation Pipeline

Used in MongoDB for data processing.

Processes documents in stages like match, group, project.

Similar to SQL GROUP BY operations.

71. Functional Programming

Programming style using pure functions.

Avoids mutable state.

Uses map, filter, reduce.
Encourages declarative coding.

72. Polymorphism

Ability of objects to take many forms.
Same method behaves differently in subclasses.
Achieved via method overriding.

73. NumPy Broadcasting

Allows operations on arrays of different shapes.
Automatically expands smaller array.
Improves vectorized performance.

74. Pandas Merge

Combines two DataFrames based on keys.
Similar to SQL JOIN.
Supports inner, left, right, outer joins.

75. Matplotlib vs Seaborn

Matplotlib is basic plotting library.
Seaborn is built on Matplotlib.
Seaborn provides advanced statistical visuals.

76. Data Cleaning Steps

Handle missing values.
Remove duplicates.
Treat outliers.
Correct data types.

77. Univariate Analysis

Analyzes one variable at a time.
Uses mean, median, variance.
Visual tools: histogram, boxplot.

78. Bivariate Analysis

Studies relationship between two variables.
Uses correlation, scatter plots.
Identifies trends and associations.

79. Central Limit Theorem

Sample mean distribution approaches normal distribution.
Applies for large sample size (≈ 30).
Regardless of population distribution.

80. Hypothesis Testing Steps

State H_0 and H_1 .
Select significance level.
Compute test statistic.
Compare p-value and conclude.

81. Type II Error

Failing to reject false null hypothesis.
Also called false negative.
Probability denoted by β .

82. Z vs T Test

Z-test used when population σ known.
T-test used when σ unknown.
T distribution used for small samples.

83. Feature Scaling

Rescales features to similar range.
Methods: Standardization, Min-Max scaling.
Improves ML performance.

84. Correlation vs Causation

Correlation shows relationship.
Causation implies one causes another.
Correlation does not imply causation.

85. Overfitting

Model learns noise in training data.
High training accuracy, low test accuracy.
Reduced using regularization.

86. Underfitting

Model too simple to capture pattern.
Poor performance on train and test data.

87. Linear Regression Assumptions

Linearity.
Independence.
Homoscedasticity.
Normality of errors.

88. Logistic Regression Intuition

Used for classification.
Applies sigmoid function.
Outputs probability between 0 and 1.

89. Entropy

Measures impurity in dataset.

$-\sum p \log p$

Used in decision trees.

90. Gini Index

Measures impurity.

$$1 - \sum p^2$$

Lower value means purer node.

91. Bagging vs Boosting

Bagging trains models independently.

Boosting trains sequentially correcting errors.

Bagging reduces variance; boosting reduces bias.

92. K-Means Steps

Choose K.

Assign points to nearest centroid.

Update centroids.

Repeat until convergence.

93. Elbow Method

Used to choose optimal K.

Plot K vs WCSS (Within-Cluster Sum of Squares).

Elbow point indicates best K.

94. Hierarchical Clustering

Builds nested clusters.

Agglomerative or divisive approach.

Visualized using dendrogram.

95. TF-IDF

Measures importance of word.

TF = term frequency.

IDF = inverse document frequency.

96. Tokenization

Breaking text into words or tokens.

First step in NLP preprocessing.

97. Activation Functions

Introduce non-linearity.

Examples: ReLU, Sigmoid, Tanh.

Enable complex learning.

98. Backpropagation

Algorithm to update neural network weights.

Uses chain rule.

Minimizes loss function.

99. CNN Convolution Layer

Applies filters to extract features.

Detects patterns like edges and textures.

Maintains spatial structure.

100. Vanishing Gradient

Gradients become very small during training.

Prevents deep networks from learning long dependencies.

Solved using LSTM, ReLU.

SECTION C – PROBLEM SOLVING (40 Questions)

Mathematics (101–115)

101. Compute $P(A|B)$ if $P(A \cap B) = 0.3$, $P(B) = 0.6$

$$P(A|B) = P(A \cap B)/P(B) = 0.5$$

102. Binomial: $n=5$, $p=0.5$, find $P(X=3)$

$$nCr * p^3 * (1-p)^2$$

$$10 * (0.5)^5 = 0.3125$$

103. Z-score if $X=80$, $\mu=70$, $\sigma=5$

$$80-70/5 = 2$$

104. Compute variance of $[2,4,6]$

$$\sum (x - \text{Mean})^2 / 3 = 8/3 = 2.67$$

105. Determinant of $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$

$$ad-bc = 8-5 = 3$$

106. Partial derivative of $f=x^2y$

$$2*x*y$$

107. Compute sigmoid(0)

$$0.5$$

108. Compute entropy for $p=0.5$

$$H(0.5) = -0.5 \log_2(0.5) - 0.5 \log_2(0.5)$$

We know:

$$\log_2(0.5) = -1$$

So,

$$H(0.5) = -0.5(-1) - 0.5(-1)$$

$$H(0.5) = 1$$

109. Find correlation meaning if $r=0.8$

Strong Positive Linear Relationship

110. Compute mean of $[1,2,3,4]$

$$(1+2+3+4)/4 = 2.5$$

111. Find gradient of $f=x^2+y^2$

Gradient of $f = x^2 + y^2$

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$
$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 2y$$
$$\boxed{\nabla f = (2x, 2y)}$$

112. If $\alpha=0.01$, update $\theta=5$, gradient=2

$$\theta_{\text{new}} = \theta - \alpha \cdot \text{gradient}$$

$$5 - 0.01 \times 2 =$$

$$5 - 0.02 = 4.98$$

113. Compute probability sum to 1 check

114. Standard error formula

Standard Error Formula

$$SE = \frac{\sigma}{\sqrt{n}}$$

If population σ unknown:

$$SE = \frac{s}{\sqrt{n}}$$

115. Bayes small example

SQL & Python (116–125)

116. Write SQL to count employees by dept

SELECT dept, COUNT(*)

FROM employees

GROUP BY dept;

117. SQL find max salary

SELECT MAX(salary) FROM employees;

118. Write LEFT JOIN example

SELECT e.name, d.dept_name

FROM employees e

LEFT JOIN departments d

```
ON e.dept_id = d.id;
```

119. MongoDB find age>30

```
db.users.find({ age: { $gt: 30 } })
```

120. Pandas groupby mean

```
df.groupby('dept')['salary'].mean()
```

121. Merge two dataframes

```
pd.merge(df1, df2, on='id')
```

122. NumPy reshape example

```
import numpy as np
```

```
a = np.array([1,2,3,4])
```

```
a.reshape(2,2)
```

123. Lambda square function

```
square = lambda x: x**2
```

124. List comprehension square

```
[x**2 for x in range(5)]
```

125. Write class example

```
class Person:
```

```
    def __init__(self, name):
```

```
        self.name = name
```

Statistics & ML (126–140)

126. One sample Z-test calculation

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

127. Two sample t-test concept

Used to compare means of two independent groups when σ unknown.

128. Compute CI

$$\bar{x} \pm Z \frac{\sigma}{\sqrt{n}}$$

129. Explain p-value interpretation

Probability of observing data as extreme as sample assuming null hypothesis true.

130. Explain logistic regression probability

$$P(y = 1) = \frac{1}{1 + e^{-z}}$$

131. Compute MSE

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

132. Identify overfitting case

High training accuracy + Low test accuracy.

133. Decision tree split logic

Chooses split with maximum Information Gain (or lowest Gini).

134. Random forest advantage

- Reduces overfitting
- Better generalization
- Uses bagging

135. K-Means assign cluster

Assign point to nearest centroid.

136. Calculate Euclidean distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

137. Compute sigmoid(2)

$$\begin{aligned} &= \frac{1}{1 + e^{-2}} \\ &= \frac{1}{1 + 0.135} = \frac{1}{1.135} = \boxed{0.88} \end{aligned}$$

138. Define regularization term

- L1: $\lambda \sum |w|$
- L2: $\lambda \sum w^2$

139. Gradient descent one step

$$\theta = \theta - \alpha \nabla J(\theta)$$

140. Explain confusion matrix

Actual\Pred	1	0
1	TP	FN
0	FP	TN

✓ 1 Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

✓ Measures overall correctness.

✓ 2 Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

✓ Out of predicted positives, how many are actually positive.

✓ 3 Recall (Sensitivity / TPR)

$$\text{Recall} = \frac{TP}{TP + FN}$$

✓ Out of actual positives, how many are correctly identified.

✓ Bonus: F1-Score

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Or directly:

$$F1 = \frac{2TP}{2TP + FP + FN}$$

DL & NLP (141–150)

141. Tokenize sentence

Input:

"I love AI"

Output:

["I", "love", "AI"]

142. Compute TF (Term Frequency)

$$TF = \frac{\text{Term Count}}{\text{Total Words}}$$

143. Explain ReLU

$$f(x) = \max(0, x)$$

144. Describe CNN architecture

Input → Conv → ReLU → Pool → FC → Output

145. Explain pooling

Reduces spatial size (Max/Avg pooling).

146. Why RNN for sequences

Because it maintains memory of previous inputs.

147. LSTM gates

- Forget Gate
- Input Gate
- Output Gate

148. Vanishing gradient math idea

Gradients shrink in deep networks due to repeated multiplication < 1.

149. Backprop update idea

$$w = w - \alpha \frac{\partial L}{\partial w}$$

150. Difference CNN vs RNN

CNN

Images

Spatial patterns

RNN

Sequences

Temporal patterns